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An explainable AI-based approach for predicting undergraduate students academic performance

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ABSTRACT

The accurate prediction of students' academic achievement has garnered considerable attention in the research community due to its importance in understanding students' progress and assisting them in achieving success. This study presents a novel approach for predicting undergraduate student's performance in the context of Bangladesh. The dataset contains 872 student records from multiple institutions. Initially the dataset was produced utilizing data-preprocessing techniques such as one-hot encoding, column remaining, and managing missing values. SMOTE (Synthetic Minority Oversampling Technique) and normalizing algorithms were employed to attain data balance and feature scaling, respectively. Afterwards, a total of seven distinct machine learning (ML) classifiers, with hyperparameter tuning, were employed to train and test in order to achieve the prediction of students' academic performance. Furthermore, a custom stacking ensemble classifier was utilized, which attained an accuracy of 86.38 %. This classifier outperformed the machine learning classifiers based on the four performance evaluation metrics. Two eXplainable Artificial Intelligence (XAI) algorithms, namely SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), were integrated to provide a comprehensible prediction of the best model and determine the significant factors. This approach provided transparency, fairness and reliability on prediction that improved student performance in the classroom and anticipation.

1. Introduction

The education system shapes a country's social and economic landscape. It serves not only as a conduit for knowledge transfer but also as an essential component of societal progress. In the last few decades, there has been a significant growth in higher education in Bangladesh [1] due to its pivotal role in developing highly skilled individuals for the country. Higher education is indispensable for advancement, as it can provide individuals with contemporary technology, skills, and knowledge [2]. The first semester of higher education serves as a transitional phase for students from a structured learning environment to an autonomous learning environment. During this stage, students develop new study habits, time management abilities, and social and psychological adjustments that influence their future academic achievement. Students' academic achievement throughout this stage may serve as a reliable indicator of their long-term results. Therefore, prediction of student performance at this stage is of paramount importance. Academic performance of students' can be influenced by a variety of factors, including socio-economic status, past academic history, present academic practices, and other personal behaviors. Timely identification of these factors affecting students' performance can help different stakeholders, such as teachers, policymakers, and educational institutions, to improve the learning experience plan by adjusting the educational policies and curriculum. By precisely predicting students' academic performance at an early stage, educational institutions may provide students additional help, such as assistance and tutoring resources [3], and instructors and teachers can utilize these prediction data to enhance their teaching-learning strategies [4]. Additionally, identifying the factors that influence students' performance, may help policymakers design a better educational system. Therefore, to ensure and maintain the quality of higher education and increase the reputation of universities, early prediction of students' performance is a must. Recognizing

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Table 1
Survey of existing studies on students' performance using ML.

Author	Sample Size	Size of Attribute	Data Source	Algorithm	Accuracy	Purpose
M. Zulfiker et al. [16]	400	8	Daffodil International University	SVM, KNN, LR, DT, AB, Multilayer Perceptron (MLP), ExtraTree, Weighted Voting Classifier	Weighted Voting Classifier - 81.73 %	● Predicted final course grades using machine learning algorithms. ● Predict student failure risk. ● Compare the performance of different machine learning algorithms.
M. Anwar et al. [17]	518	17	Daffodil International University, academic year 2019	ANN, KNN, GNB, RF, DT	ANN- 88 %	● Predicted students' performance for determining their category and enabling them to precisely identify their capacity for learning.
F. Janan et al. [18]	583	21	Bangladesh University of Textiles	SVM	SVM- 81.25 %	● Predicted students' distinct levels of outcomes based on the changed survey data, which has been modified using fuzzy logic.
M. Ashaduzzaman et al. [19]	687 students with 38681 instances	7	Green University of Bangladesh	Naïve Bayes, J48 Decision Tree, SVM	SVM- 89.1262 %	● Used data mining techniques to determine the academic factors that affect final grades, guiding educational strategies.
K. Ahammad et al. [20]	400	31	Feni Model High School, Bangladesh	Naïve Bayes, SVM, KNN, XG-boost, MLP	MLP- 86.25 %	● Predict student performance to identify at-risk students.
M. Hoque et al. [21]	850	14	NM	J48, REPTree, Hoeffding Tree	J48–85.6471 %	● To help students forecast and improve academic achievements ● Provide suggestions for present and future students ● Predicts grade points using decision tree algorithms, specifically the J48 algorithm

the significance of predicting undergraduate students' academic achievement, several countries have already conducted research in this area [5–7].

In the last few decades, machine learning (ML) methods have been used to forecast students' academic performance based on demographic information, current academic practices, socio-economic factors, and prior academic information [8]. The objective of using these machine learning techniques in predicting students' academic achievement is to automate the prognosis of student's outcomes [9]. But the black-box nature of these models makes them hard to trust. Nevertheless, comprehending the forecasts produced by these machine learning methods is of utmost importance when handling human data [10]. This highlights the need of explainable artificial intelligence (XAI), particularly in the realm of education. An explainable model facilitates comprehension and engenders confidence in forecasts for instructors, students, policy-makers, and other stakeholders. It also assists educators in adjusting their pedagogical approaches and offering focused assistance to at-risk students. Explainable performance prediction models in education have previously been the subject of a few studies [11–13]. Based on our review of the literature, we have not found any research that utilizes an explainable model to assess the academic performance of undergraduate students in a Bangladeshi educational setting based on semester outcomes. This motivated us to make a contribution in this area.

In this study, we proposed an explainable stacking ensemble model for predicting undergraduate students' semester results in the context of Bangladesh. Prediction of semester results is challenging as it may vary from person to person due to various factors. In order to tackle this issue, we gathered data from undergraduate students hailing from several institutions in Bangladesh, with a specific emphasis on identifying characteristics that might potentially influence their academic performance. To prepare the data, a number of preprocessing methods were used, including column renaming, handling missing values, and one-hot encoding. Moreover, the SMOTE method was used to balance all classes and StandardScalar was utilized to scale all the features, which greatly improved the accuracy of our model. The stacking ensemble model consists of Random Forest (RF) and Gradient Boost (GB) as base classifiers, with Support Vector Classifier (SVC) serving as the meta-classifier. We conducted a performance comparison between our custom model and numerous baseline classifiers, including Random Forest (RF), XGBClassifier (XGB), Gradient Boosting (GB), Support Vector Classifier (SVC), K-Nearest Neighbors (KNN), Decision Tree (DT), and Logistic Regression (LR). The experimental findings indicated that the stacking ensemble model significantly outperformed these baseline algorithms. For interpretability of the model's output, we incorporated XAI techniques such as LIME and SHAPE [14]. The significance of each feature is calculated by SHAP values, which helps to comprehend the influence of each feature on models' output [15]. While the LIME technique helps in understanding the model's output locally by highlighting the most significant features of a specific student's predicted performance. The main contribution of this paper includes.

- Insight into the most significant feature regarding student performance in developing a system.
- Investigation and prediction of students' academic performance at the initial phase of their undergraduate life, which facilitates higher education journeys.
- Developed an ensemble model based on two classifiers that can forecast students' results more accurately.
- Integration of explainable artificial intelligence techniques such as SHAP and LIME to enhance model interpretability and provide insightful explanations.

The subsequent sections of the article are structured in the following manner: Section 2 provides an overview of the existing literature, while Section 3 outlines the methods used in the study. Section 4 presents the experimental results, compares them with the existing literature, and discusses the findings. Finally, Section 5 concludes the study and suggests areas for further research.

2. Literature review

In order to enhance our comprehension of the context, we have partitioned the literature into two distinct sections. Initially, we have analyzed the research that predicted the academic achievement of undergraduate students within the specific setting of Bangladesh. This section examines studies on several factors that impact students'

Table 2
Survey of existing studies on students' performance prediction using XAI.

Author	Year	Dataset	Purpose	XAI Method	Algorithm	Highest Accuracy
I. Livieris et al. [23]	2023	DATASET1(337) DATASET2(3716)	Present an advanced framework for precise, interpretable prediction of students' performance.	LIME SHAP	LGBM, Random Forest, XGBoost, NGBoost	NGBoost 90.0 %
S. ALWARTHAN et al. [24]	2022	Collected from Imam Abdulrahman bin Faisal University	Early detection of students who are most likely to fail in the preparation exam.	LIME SHAP	SVM, RF, ANN	RF-99.662 %
al. [25M. Afzaal et al. [25]	2023	585 records with six features	Proposed an XAI approach that provides automatic feedback and recommendations on students' learning.	Counterfactual Explanation (CF)	RF, ANN, KNN, SVM, BayesNe, LR	- RF 88 % - ANN 88 %
M. Nagy et al. [26]	2023	8508 records with eleven features	Deploy explainable AI techniques to deal with student dropout in STEM-related fields.	LIME SHAP	CatBoost, NGBoost, Logistic Regression, XGBoost, Gradient Boosting, AdaBoost, Light Gradient Boosting	- CatBoost 77.4 %
H. Chen et al. [27]	2022	Dataset consist More than 202,000 logs of 977 students collected from Gadjah Mada University in Indonesia.	Develop a system for predicting student achievement in online courses during COVID-19.	LIME	LR SVM LSTM	CNN-LSTM −91 %
M. ADNAN et al. [28]	2022	32,593 records	Propose an XAI-based model for interpreting students' performance in various course lengths.	LIME SHAP	LR, Stochastic Gradient Descent (SGD), Naïve Bayes (NB), KNN, DT, SVC, Bagging Classifier, RF, Extra Tree, Gradient Boosting, AdaBoost, Voting Classifier.	Random Forest- 95.5 %

academic achievement in educational environments. Furthermore, we have analyzed studies that have used XAI methods to elucidate models in the prediction of student performance.

M. Zulfiker et al. [16] used machine learning models to predict students' grades at private institutions in Bangladesh. For weighted vote aggregation, they obtained an accuracy of 81.73 %. The main motive of this study was to help students improve their grades and support teachers in detecting at-risk students. The limitation of this study is that they did not compare the performance of their proposed method with existing studies. **M. Anwar et al**. [17] suggested a process of web-based four-tier architecture for student performance prediction, which identified the categories of students to assist them in identifying their learning capacities. The system includes data collection, preprocessing, model training, and testing and focuses on feature importance by using an artificial neural network (ANN) model. However, the KNN model achieved a very lower accuracy, which is 37 % on their dataset. **F. Janan et al**. [18] utilized SVM to categorize students' various performance levels and predict their results. The input data for training and testing were blended from surveys of students and experts using an adaptive neuro-fuzzy interference system (ANFIS). The suggested model's applicability for forecasting course-wise performance might be improved by including additional features and expanding survey participants, resulting in higher accuracy. However, there is a clear gap in their dataset, as the dataset is limited. **M. Ashaduzzaman et al**. [19] used a data mining approach to determine the academic factors that affect students' final grades or CGPA. Three Ml algorithm decision tree, Naïve Bayes, and SVM were used to predict students' performance. The constraint of this research lies in its limited applicability, as it only pertains to a certain private institution in Bangladesh. This is due to the fact that each private university employs distinct methods and strategies to evaluate its students.

Again another study, **K. Ahammad et al**. [20] predicted students' performance at the secondary level to identify at-risk students. The data was collected from Fani Model High School, Bangladesh. The limitation of this study is that their data collection process covered only a single school, which limits the generalizability. **M. Hoque et al.** [21] presented a web-based system that predicts university students' grades and identifies areas for improvement based on previous data and questionnaires, utilizing the J48 decision tree algorithm. It proves the better accuracy of J48 by comparing decision tree algorithms. The study's drawback is that, although other variables like financial circumstances could have an influence on students' performance, the researchers exclusively used academic features to predict students' achievement. **M. Mia et al.** [22] utilized more than a thousand data to predict student retention in a private university. They utilized SVM, Naive Bayes, Logistic, JRip, J48, Multilayer Perceptron, and Random Forest classifiers and among them, SVM outperformed the other classifiers with an accuracy of 85.76 %. Although this research obtained a high level of accuracy, it became limited by the use of a small number of features, which is its main drawback. The above literature reviews considering Bangladeshi student's performance predictions are also summarized on Table 1.

I. Livieris et al. [23] suggested a framework that emphasized the interpretability and explainability of the models for predicting students' performance. To provide accurate, dependable, and understandable results, the framework uses the NGBoost algorithm for prediction and the LIME and SHAP methods for local and global explanations. The main drawback of this study is that it has used two real-time datasets, however, both datasets have a restricted range of features. **S. ALWARTHAN et al. [24]** identified at-risk students who are most likely to fail in preparatory exams at a University IAU using RF (Random Forest), ANN, and SVM algorithms. For model explainability, LIME and SHAP are integrated in this study. The primary constraint of this research is that the dataset was obtained via an automated method; hence, some attributes including personality, parent education, and parent occupation were not considered for predicting performance, thereby restricting the study's generalizability. **M. Afzaal et al. [25]** developed an Explainable AI-based dashboard to improve students' ability and control their own learning. They utilized RF, ANN, KNN, SVM, BayesNe, and LR classifiers. RF and ANN achieved an accuracy of 88 %. For explainability the utilized Counterfactual Explanation (CF). An inherent limitation of this research is its exclusive focus on a single programming course, hence restricting its applicability to a broader context. **M. Nagy et al. [26]** emphasized personalized interventions by predicting dropout in higher education using interpretable machine learning and explainable AI tools. It employed approaches such as permutation significance, partial dependence plots, LIME, and SHAP values to provide both global and local explanations of predictions. Nevertheless, this research relies on the data from Budapest University of Technology and Economics, where the majority of students are pursuing STEM degrees. Consequently, it may be necessary to conduct further analysis using data from other higher education institutions.

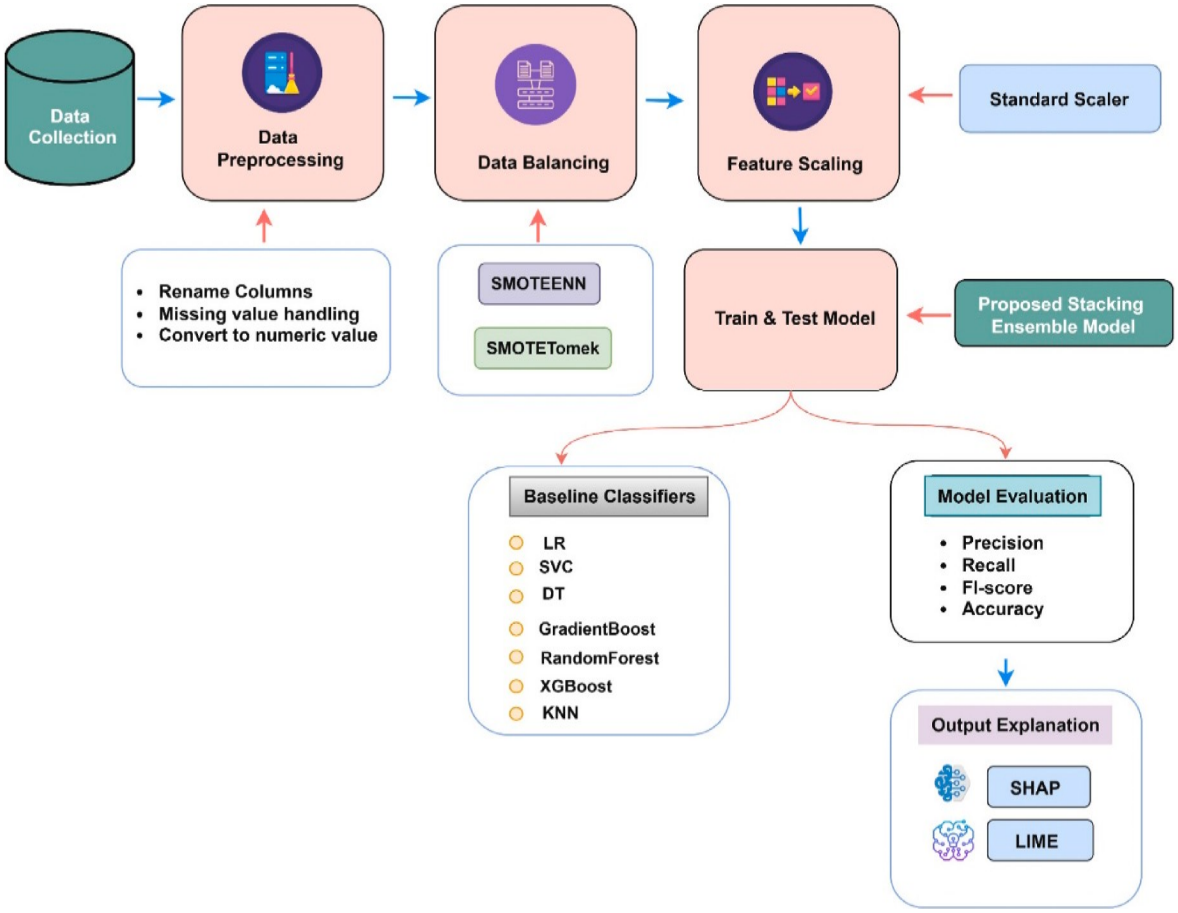


Fig. 1. Working Procedure of Student Performance Prediction using XAI.

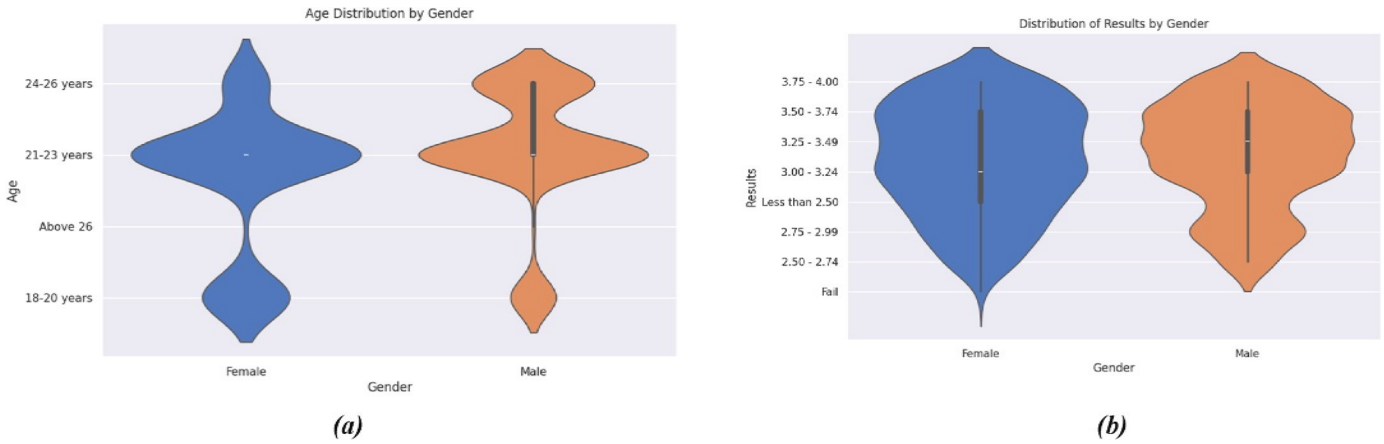


Fig. 2. Visualization of (a) Students Age Distribution by Gender. (b) Student's Academic Result Distribution based on Gender.

Another study by **H. Chen et al. [27]** presented an intelligent framework for early student performance prediction in virtual learning environments during COVID-19, based on hybrid data sampling CNN-LSTM and Conv-LSTM models that beat LSTM, SVM, and LR baseline models with F1-scores of 0.91 and 0.88, 0.84, 0.79, and 0.64, respectively. Nevertheless, the author suggested the potential expansion of this research to examine the students' performance when the instructor intervenes during learning activities. **M. ADNAN et al.** [28] focused on predicting students' performance using machine learning and explainable artificial intelligence (XAI) in an virtual learning environments (VLEs) at different stages of the course (20 %, 40 %, 60 %, 80 %, and 100 %). However, the comprehensiveness of this study's analysis may be restricted. As this research does not thoroughly investigate the interaction or contribution of various types of factors (demographic, clickstream, assessment) to the model's predictions over time, it fails to give deeper insights into student behavior. The overall summary of applying XAI techniques to predict student performance is presented in Table 2.

From the above analysis, we can claim that most of these studies that are conducted in Bangladesh or international scope gather datasets from a single institution or department, which actually limits the generalizability of their findings. Furthermore, many studies have focused on predicting the performance of specific courses and only limited studies

Table 3
Comprehensive description of the utilized dataset.

Serial	Feature	Description	Serial	Feature	Description
1	Student's Age	Represents participants' age.	16	Scholarship in SSC	Whether students got scholarships during their SSC.
2	Gender	Refers to a student's sexual identity	17	Scholarship in HSC	Whether students got scholarships during their HSC
3	Relationship Status	The participant's relationship condition, whether he/she is "In a love relationship", "single", or "married".	18	Accommodation	student's living condition, situation, or surroundings.
4	Type of University	The type of university the participants are studying at.	19	Weekly Study Time at Home	Weekly study hours of participants at home which provide insights into students' study habits, time management abilities, and dedication.
5	Father's Education	Represents father's educational qualification.	20	Attendance in Class	Represents the regular attendance percentage of a student in the class.
6	Mother's Education	Mother's educational qualification.	21	Mid-Term Exam Preparation	How a student prepared for their midterm exam and their study habits
7	Father's Occupation	Denotes the work or job done by a participant's father for livelihood.	22	Taking Exam Preparation	Provides insight about students' exam preparation techniques such as regular study during the semester', 'Close date to the exam', and 'Never'.
8	Mother's Occupation	Refers to a participant's mother's occupation and is also classified as similar to a father's occupation.	23	Reading Scientific Books, Articles, and Journals.	States that the student has a habit of reading scientific books, articles, and journals, or not.
9	Family Income	The monthly family income of a student. It provides information about students' family income, financial condition, and socioeconomic origins.	24	Taking Class Note	Information about student note-taking behavior in the classroom.
10	Number of Siblings	The number of siblings the participants have.	25	Any Co-curricular Activity	Reflects whether students participate in extracurricular activities or not.
11	Parental Status	Participants' parents' marital status.	26	Attend any Seminar Related to the Department	Represents whether the participant participates in departmental events and academic enrichment programs or not.
12	Previous School Location	Previous educational location of the student.	27	Reading Non-Scientific Books, Articles, and Journals.	States that the student has a habit of reading non-scientific books, articles, and journals, or not.
13	Previous College Location	Information about participants' previous college location. It may shed light on how educational opportunities and experiences might change or stay the same when students move from college to university.	28	Any part-time job	If the participant is doing any other part-time job or work besides his academic study.
14	SSC result	Students' academic achievements during secondary school, offering a key indicator of their educational performance.	29	Smoking	It captures the student's smoking behavior and offers information about their tobacco or cigarette smoking trends or not.
15	HSC result	Higher Secondary Certificate (HSC) results, provide insights into their academic performance and affect their educational and future possibilities.	30	Sports/Exercise	Indicates how frequently the student participates in sports or physical exercise.

have focused on semester results. Additionally, these studies used machine learning models for prediction but did not explain the model outputs, which limits models' interpretability. These gaps in the existing literature motivated us to contribute to this sector. This study aims to address these issues by predicting students' performance at an early stage by using interpretable ML algorithms.

3. Material and methods

This section outlines the comprehensive research method used to accomplish this study's objectives. The overall working procedure of our study is depicted in Fig. 1. Additionally, the following subsections expound upon the main phases of the suggested model.

3.1. Data description

The dataset used in this study was obtained from students ranging in age from 18 to 26 who are currently enrolled in different universities using Google Forms. Stringent ethical norms were adhered to throughout the data-gathering procedure to ensure respondent confidentiality, privacy, and informed consent. The dataset comprises 34 columns in total and three columns were excluded during preprocessing, whereas one column represented date-time information, another contained respondent names, which were omitted to safeguard anonymity and another column was about the department name, which was collected as an open-ended response. Finally, we utilized 31 columns, where 30 for the prediction and 1 column as our target. For further details and reproducibility, the complete dataset is publicly available on GitHub Some data visualization is presented in Fig. 2.

All 30 columns consist of categorical values that influence students' academic results. The features of the dataset are described below Table 3.

3.2. Data pre-processing

Data preparation is essential for tackling missing values, uneven classifications, and a lack of patterns. Firstly, the obtained dataset was checked to see whether there were any missing and null values or not. Nevertheless, several column names were too long due to the fact that the data was collected via Google Form. In order to enhance legibility, the columns in the dataset were renamed. Table 4 provides a comprehensive summary of the dataset after decoding it to numerical data.

Table 4
Utilization of feature encoding technique on preprocessed data.

Serial	Feature Name	Renamed Feature	Feature Encoding	Serial	Feature Name	Renamed Feature	Feature Encoding
1	Students Age	Age	18–20 years = 1 21–23 years = 2 24–26 years = 3 Above 26 = 4	**16**	Reading Scientific books/ articles/journals	Scin_reading	No = 0 Yes = 1
2	Gender	Gender	Female = 0 Male = 1	**17**	Scholarship in HSC	HSC_schlr	No = 0 Yes = 1
3	Relationship status	R_status	Single = 0 In a love relationship = 1 Married = 2	**18**	Attendance in class	Cls_attendence	Less than 60 % classes = 0 60 %–79 % classes = 1 More than 80 % class = 2
4	Type of University	Uni_type	Others = 0 Private = 1 Public = 2	**19**	Mid-Term Exam Preparation	Mid_prep	Not applicable = 0 Alone = 1 Group study = 2
5	Father's education	Fa_Edu	Primary School = 0 Secondary School = 1 College = 2 B.A./B.Sc./Hons/ Degree = 3 M.Sc/M.A. = 4 'Ph.D. = 5	**20**	Weekly Study Time at Home	Stdy_time	Less than 1 h = 0 2–5 h = 1 6–10 h = 2 11–14 h = 3 11–15 h = 4 More than 15 h = 5
6	Mother's education	Mo_Edu	Primary School = 0 Secondary School = 1 College = 2 B.A./B.Sc./Hons/ Degree = 3 M.Sc/M.A. = 4 'Ph.D. = 5	**21**	Smoking	Smoking	No = 0 Yes = 1
7	Father's Occupation	Fa_Occu	Retired = 0 Business/Self-employed = 1 Private Job = 2 Govt. Service = 3	**22**	Sports/Exercise	Sprts_Exer	Never = 0 Sometimes = 1 Always = 2
8	Mother's Occupation	Mo_Occu	Housewife = 0 Business/Self-employed = 1 Private Job = 2 Govt. Service = 3	**23**	HSC result	HSC_rslt	Less than 3.00 = 0 GPA 3.00–3.99 = 1 GPA 4.00–4.99 = 2 GPA 5.00 = 3
9	Family Income (monthly)	Mnthly_Income	<25,000Tk = 0 26,000–40,000 TK = 1 41000-55000 TK = 2 56000-60,000 = 3 60,000 = 4	**24**	SSC result	SSC_rslt	Less than 3.00 = 0 GPA 3.00–3.99 = 1 GPA 4.00–4.99 = 2 GPA 5.00 = 3
10	Number of Siblings	Siblings	No Siblings = 0 1 = 1 2 = 2 3 or above = 3	**25**	Any part-time job	Part_time_job	Not Applicable = 0 Tution = 1 Others = 2 Business = 3
11	Parental status	P_Status	Divorced = 0 Happily Married = 1 Died-one of them or both = 2	**26**	Accommodation	Accom	Others = 0 With family = 1 Rental = 2 University hall = 3
12	Previous School Location	Pre_scl	Rural = 0 Urban = 1 Country Town = 2	**27**	Taking Exam Preparation	Exam_prep	Never = 0 Close date to exam = 1 Regular study during the semester = 2
13	Previous College Location	Pre_clg	Rural = 0 Urban = 1 Country Town = 2	**28**	Taking Class Note	Cls_Note	Never = 0 Sometimes = 1 Always = 2
14	Reading Non-Scientific books/articles/journals	Non_Scin_reading	No = 0 Yes = 1	**29**	Attend any Seminar related to department	Seminar	No = 0 Yes = 1
15	Scholarship in SSC	SSC_Schlr	No = 0 Yes = 1	**30**	Any Co-curricular activity	Co_activity	No = 0 Yes = 1

Table 5
Dataset distribution based on the target column.

Result (CGPA)	Class	Number of Participants	Percentage
3.75–4.00	Excellent	116	13.30 %
3.25–3.74	Good	348	39.91 %
2.75–3.24	Average	275	31.54 %
Less than 2.74	Below Average	133	15.25 %

The result column was categorized into four classes and these categories are actually aligned with common university practices in Bangladesh. Also, Label encoding was done for the target column named "Results, where 'Below Average': 0, 'Average': 1, 'Good': 2, 'Excellent': 3. The overall obtained dataset distribution considering the target column is presented in Table 5.

3.2.1. Data balancing

There was an imbalance in our dataset once the target column was divided into four groups. To address this issue, we employed data

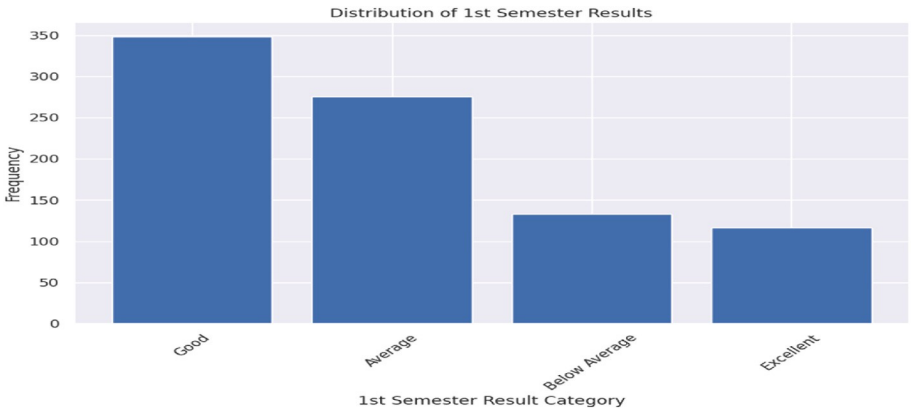


Fig. 3. Visualization of data before applying SMOTE technique.

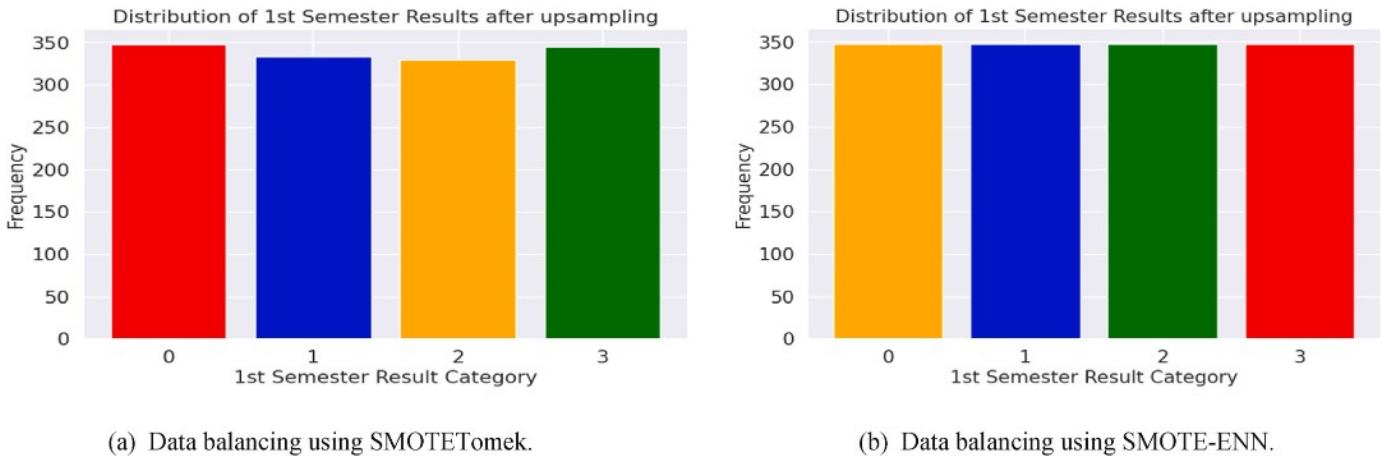


Fig. 4. Visualization of data after applying SMOTE techniques.

balancing methods to ensure that the model training process remained unbiased toward any particular class. Two primary techniques are commonly employed: oversampling and undersampling. Undersampling involves reducing the size of the majority class, which can enhance user-friendliness and decrease runtime. However, this method also diminishes the number of data points available for analysis. As our dataset's already limited size of 872 data points, this study did not utilize undersampling. Rather we chose Oversampling to address the imbalance and guarantee that every class was fairly represented. Oversampling indeed involves generating synthetic data points from the minority class to balance the dataset [29]. For this dataset, to mitigate the risk of inappropriate model training due to class imbalance, the Borderline-Synthetic Minority Oversampling Technique (SMOTE) is employed [30]. From the documentation of SMOTE, for resampling the categorical value SMOTEN can be used [31]. Also, for comparison of model output, we utilized SMOTETomek which is a combination of over and under-sampling methods [32]. SMOTEN identifies the k-nearest neighbors of a minority class sample, from which a new synthetic sample is created by selecting at random attribute values from one of the k-nearest neighbors. It guarantees that the novel synthetic data preserves a plausible category distribution. Conversely, SMOTETomek first use SMOTE to create synthetic samples for the minority class, thereafter eliminating Tomek linkages which are pairings of proximate closest neighbors from different classes. SMOTETomek balances the dataset by deleting redundant majority class samples, therefore refining class borders, lowering noise, and enhancing model performance. Figs. 3 and 4 illustrates a comparison between imbalanced and balanced datasets, showcasing the distribution of data points across different classes before and after applying data balancing techniques.

3.3. *Feature scaling*

Feature scaling, sometimes known as feature normalization, is a technique used to ensure that all features in a dataset are on the same scale. It is used to ensure that the feature being used is treated in the same manner. To increase the model accuracy on a broad scale we used the StandardScaler technique [33].

3.4. *Splitting data into train and test*

After completing the pre-processing steps, we divided the dataset into 2 parts for training and testing. The train set consists 80 % while the test set consists of 20 % of the dataset.

3.5. *Machine learning models and XAI terminologies*

We employed seven machine learning classifiers, encompassing Random Forest, XGBClassifier, Gradient Boosting, Support Vector Classifier (SVC), K-Nearest Neighbors (KNeighbors), Decision Tree, and Logistic Regression. To eliminate the black-box nature of the ML algorithm, two XAI approaches named SHAP and LIME are utilized.

3.5.1. *Random Forest (RF)*

Random Forest (RF) is an ensemble learning method that enhances the performance of individual decision trees by employing a technique called bootstrap aggregating (or bagging), where multiple trees are

trained on random subsets of the data and features [34]. The fundamental principle behind it is to ensemble a collection of individual decision trees, each trained on a random subset of the data and features [35].

Gini index or entropy are frequently used by RF to solve classification difficulties. Here, we've hypertuned RF model using the entropy and Gini indexes. The formula for the Gini index is:

$$Gini = 1 - \sum_{i=1}^{c} (P_i)^2 \tag{1}$$

Where P_i = Relative Frequency;

Unlike the Gini index, Entropy is more mathematical intensive as it contains a logarithmic function. For entropy, the equation is:

$$\text{Entropy} = \sum_{i=1}^{c} - P_i * \log_2 (P_i)^2 \tag{2}$$

3.5.2. Gradient Boosting(GB)

GB is another technique for ensemble learning is gradient boosting, which builds trees one after the other, fixing any errors of the previous one. It works based on three main concepts: loss function, weak learner, and additive model. For classification tasks, it mainly works on sequentially building weak learners. For multiclass classification, the model uses the following function for predicting the probabilities of each class:

$$P_k(x) = \frac{e^{F_{K,M^{(x)}}}}{\sum_{j=1}^{k} e^{F_{i,M^{(x)}}}} \tag{3}$$

Where,

$F_{K,M^{(x)}}$ = the final model for class k after M iteration;

$P_k(x)$ = Prediction probability

3.5.3. XGBClassifier (XGB)

XGBClassifier, a prominent supervised learning algorithm, is a variant of the gradient boosting algorithm known for its efficiency and effectiveness in handling complex datasets. To overcome the difficulties of GB, an improved version called XGBoost was developed, providing improved performance and scalability for training even on large datasets [36].

XGBBoost method is built on gradient boosting trees, therefore it is more useful for gradient enhancement [37]. For dataset D = $\{(x_i, y_i)\}$ that contains m features and n samples, the result of sample prediction can is represented by the following equation:

$$\hat{y_i} = \sum_{k=1}^{k} f_k(x_i), f_k \in \varphi \tag{4}$$

$$\varphi = \{f(x) = \omega_s(x)\} \ \left(s : R^m \rightarrow \ T, \omega_s \in R^T\right) \tag{5}$$

Where,

$\hat{y}_i$ = prediction label;

x_i = each ith samples

$f_k^{(x_i)}$ = predicted score for the samples

$f(x)$ = leave's value

ω = number of leaves

3.5.4. Support Vector Classifier (SVC)

SVC (Support Vector Classifier) is a supervised learning method based on the SVM (Support Vector Machine). It focuses on identifying the ideal hyperplane for separating the classes. By identifying the best hyperplanes, SVC gives valuable insights for decision-making and resolving problems.

SVM utilizes a set of mathematical functions called "kernel". We have utilized two kernels 'rfb' and 'linear'. For linear kernel, the equation is:

$$K(x, y) = sum\ (x, y) = \sum_{i=1}^{n} x_i y_i \tag{6}$$

Where,

X = (x_1, x_2, …. …., x_n) represents an n-dimensional input vector;

Y = (y_1, y_2, …. …., y_n) represents another n-dimensional input vector;

x_i and y_i represents i^{th} component of x and y correspondingly;

n = dimensionality of the input vector.

Again, for rfb kernel, the following equation is utilized:

$$\text{k}\ (\text{x}, \text{y}) = e^{-\frac{(||x-y||)}{\sigma}} \tag{7}$$

Where,

σ = variance

||x - y|| = Euclidean Distance between two points x and y.

3.5.5. K-Nearest Neighbors (KNN)

KNN is a non-parametric supervised learning classifier that can be used for regression and classification applications. It calculates the distance between the test and training data points during prediction, then uses this distance metric to choose the K nearest neighbors. The right class or regression value for the test data is subsequently ascertained using these nearest neighbors [38]. The distance metrics can be calculated by various methods such as Minkowski distance, Manhattan distance etc. For Manhatten distance metric formula:

$$Manhatten\ Distance = sum\ for\ i\ to\ N\ sum\ \|x_1 - x_2\| \tag{8}$$

Where,

x_1, x_2 = 2 − dimensional vector

Again, For Minkowski distance the formula is represented by:

$$\left\|x_1 - x_2\right\| = \left(\sum_{i=1}^{d} |x_{1i} - x_{2i}|^p\right)^{y^p} \tag{9}$$

3.5.6. Decision Tree (DT)

DT is a method for supervised learning that works in a similar way to the human mind. Every internal node in DT represents a "decision" made in response to a feature, and every branch in DT shows the decision's results. The final decision or prediction is then represented by the leaf node. The model performs better the deeper the tree and the more branches it has [39]. These can be solved by using ASM (**Attribute selection measure**) such as gini index and entropy. The formula for Gini index is:

$$Gini = 1 - \sum_{i=1}^{n} (p_i)^2 \tag{10}$$

Again, the entropy is represented by the formula:

$$Entropy = \ - \sum_{i=1}^{n} p_i\ log_2\ (p_i) \tag{11}$$

Where, p_i = Probability

3.5.7. Logistic Regression (LR)

When used for binary classification, logistic regression is a statistical method that performs well, especially for classes that are linearly separable. However, the model may be expanded to include multiclass classification situations using multinomial logistic regression or softmax regression methods [40]. The formula that represents LR is:

$$y = \frac{e^{(b_0 + b_1 x)}}{1 + e^{(b_0 + b_1 x)}} \tag{12}$$

Where,

X = input value.

Table 6
Hyperparameters grid used for grid search of RF, GB, XGB, SVC, KNN, DT, LR classifier.

Classifier name	Parameters	Value
RF	n_estimators	[50,100,200]
	criterion	['gini', 'entropy']
	max_depth	[None, 10, 20, 30]
	min_samples_split	[2,5,10]
	min_samples_leaf	[1,2,4]
GB	n_estimators	[50,100,200]
	learning _rate	[0.01, 0.1, 0.5]
	max_depth	[3,5,7]
XGB	n_estimators	[50,100,200]
	learning _rate	[0.01, 0.1, 0.5]
	max_depth	[3,5,7]
SVC	C	[0.1, 1, 10, 100, 300]
	gamma	[0.1, 0.05, 0.01, 0.001]
	kernel	['rbf', 'linear']
KNN	n_neighbors	[3,5,7]
	weights	['uniform', 'distance']
	metric	['euclidean', 'manhattan']
DT	criterion	['gini', 'entropy']
	max_depth	[None, 10, 20, 30, 40, 50, 100]
	min_samples_split	[2,5,10,20]
	min_samples_leaf	[1,2,4,5]
LR	C	[0.1, 1, 10, 100]
	penalty	[11,12]

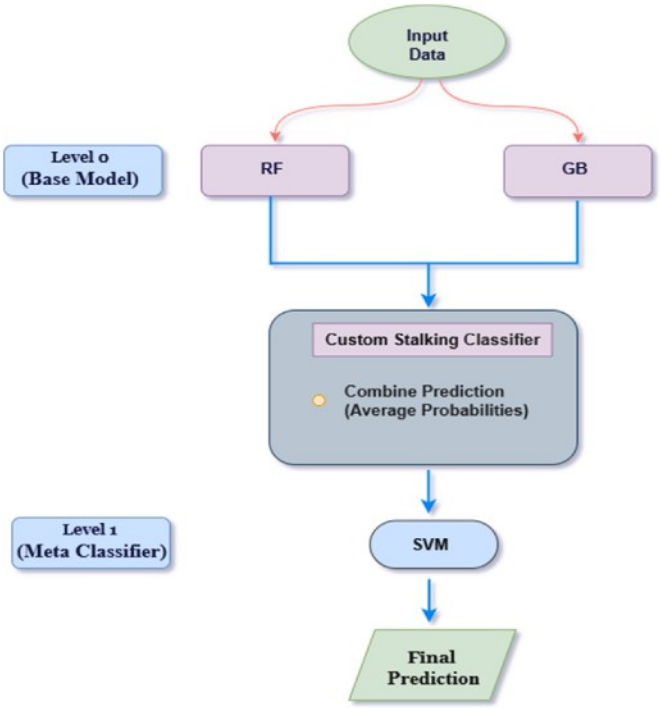


Fig. 5. Working procedure of proposed stacking ensemble model.

Y = predicted output.
b_0 = intercepted term or bias
b_1 = coefficient for x

Below Table 6 describes the hyperparameter grid we used for our baseline models.

3.5.8. *Proposed stacking ensemble model*

To predict students' performance more accurately, we designed a custom stalking ensemble classifier. This classifies as a combination of both bagging and boosting. Our proposed stalking classifier consists of two base models RF and GB. As the meta model, the SVC was utilized to enhance the model's robustness and improve predictive accuracy. The base models and meta-models are described below:

- **Base Models (Level-0 Models):**

The base model RF reduces the variance by training multiple decision trees on randomly bootstrapped samples. Also, RF has shown efficacy in high-dimensional areas and has been effectively used in unbalanced tasks [41]. On the other hand, the GB reduces bias by sequentially improving weak learners. It focuses on correcting the mistakes made by previous models. By assigning higher weights to misclassified samples, GB reduces bias and improves prediction accuracy. The Gradient boosting trees (GBT) are supervised machine learning techniques that may represent intricate nonlinear feature interactions without necessitating assumptions about data distribution. The method enhances its predicting accuracy by progressively decreasing residual errors from preceding trees [42]. These are the reasons we chose RF and GB for our base model.

- **Meta-Model (Level-1 Model):**

In order to anticipate the final prediction, the meta-model aggregates the predictions from the base models and uses them as input features. We used SVC as a meta-model because it can identify the best decision limits in a high-dimensional feature space. It utilizes the probabilities of RF and GB and provides the final prediction.

The architecture of the stalking ensemble model is presented in Fig. 5 illustrating that the model begins by independently training the base models RF and GB independently with best parameters. Let $h_1(x), h_2(x), \ldots, h_k(x)$ us represent the predictions of k-base models. These predictions collectively form a new feature matrix Z, where Z = $[h_1(x), h_2(x), \ldots, h_k(x)]$. This transformed feature matrix Z is then passed to the meta-model SVC for final decision-making. To enhance prediction reliability we configured SVC with probability estimation enabled (probability = True), ensuring that the final model produces soft predictions using class probabilities rather than hard labels. The meta-model then integrates these probability distributions and generates the final classification outcome based on the following equation:

$$\widehat{x}_i = \int_{meta} \left(\widehat{x}_i^{(1)}, \widehat{x}_i^{(2)}, \ldots\ldots, \widehat{x}_i^{(k)}\right) \tag{13}$$

Where, $\widehat{x}_i$ is the prediction by k^{th} base model for i^{th} data point.

3.5.9. *SHAP (SHapley Additive Explanations)*

The SHAP method is developed based on game theory [43]. SHAP values can aid in understanding and identifying the factors behind students' performance as teachers, practitioners and policymakers may struggle to grasp the predictions made by the models. It is model-agnostic, which means it can describe any machine-learning model.

SHAP method simply analyzes the outputs of ML models to discover how much each attribute influences the model's output. The use of SHAP can remove the black box nature of ML models. SHAP can provide both global (entire dataset) and local (single instance) explanation by assigning SHAP values for each feature. The contribution of each feature is determined by the following formula:

$$k_i = \sum_{\zeta \subseteq G} \frac{|\zeta|!(n-|\zeta|-1)!}{n!} [\tau(\zeta \cup \{i\}) - \tau(\zeta)] \tag{14}$$

Where,
$k_i = i^{th}$ feature
G = Collection of all the features
n = number of features in G.
ζ = subset of G
$\tau(\zeta \cup \{i\})$ = mel's output with i^{th} feature

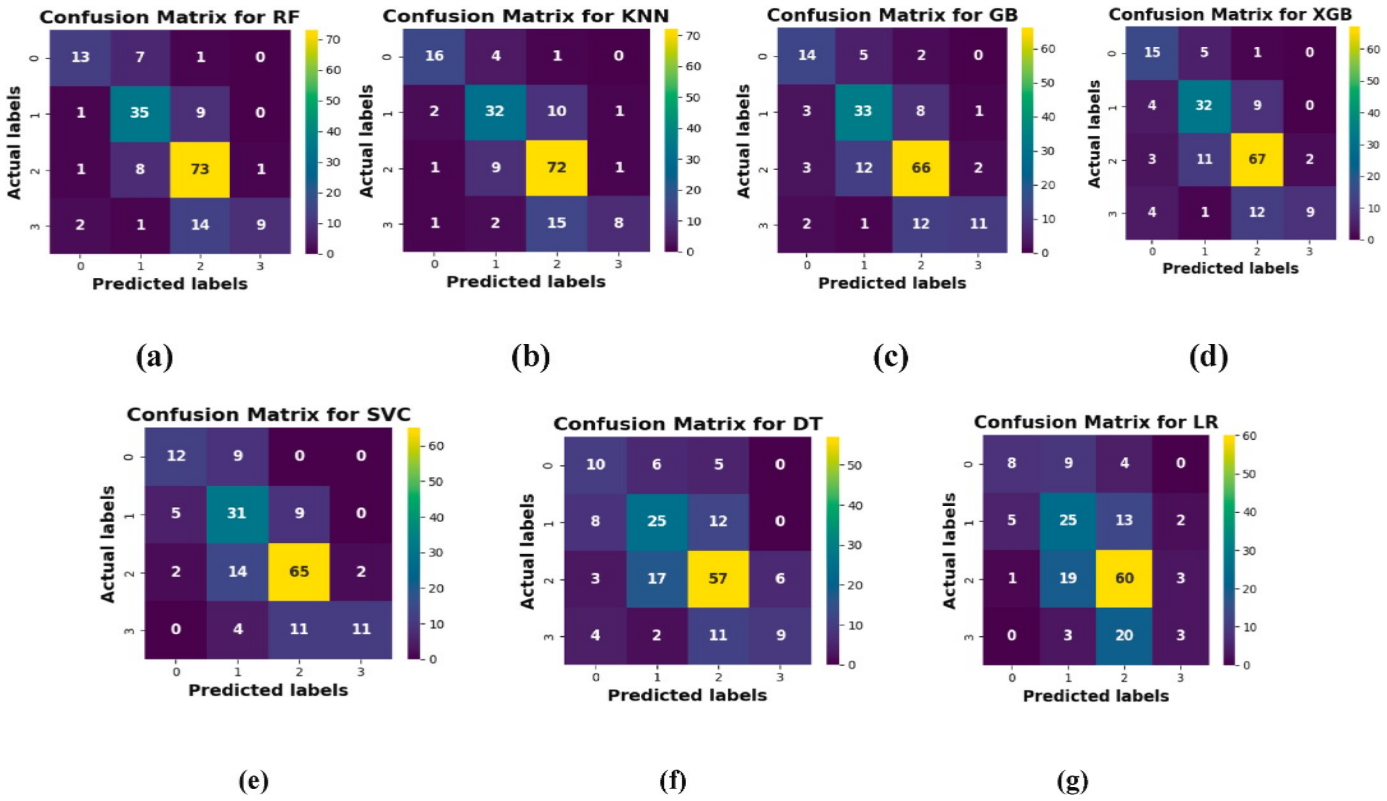


Fig. 6. 4x4 Confusion Matrix for ML classifier ((a)RF (b)KNN (c)GB (d)XGB (e)SVC (f)DT (g)LR) Before Applying Scaling Techniques.

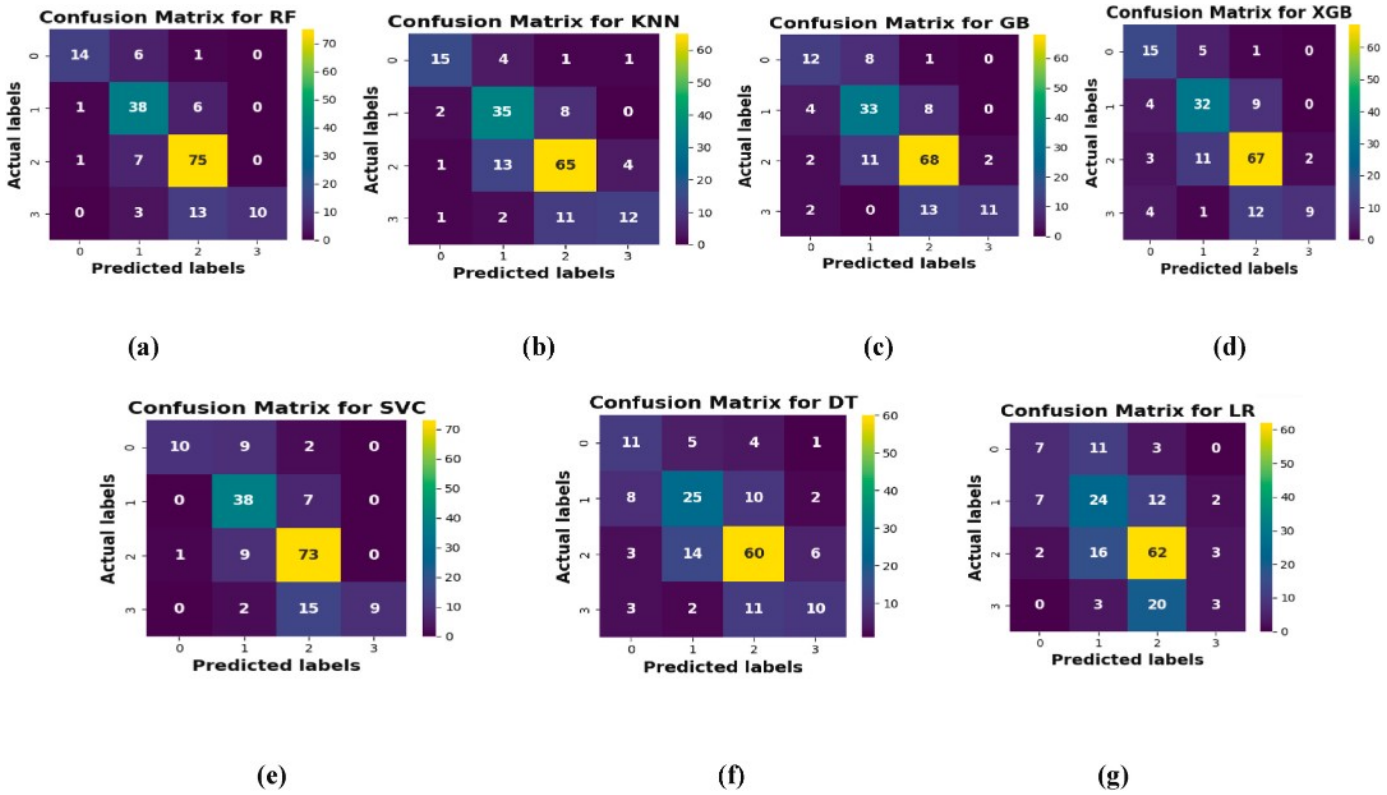


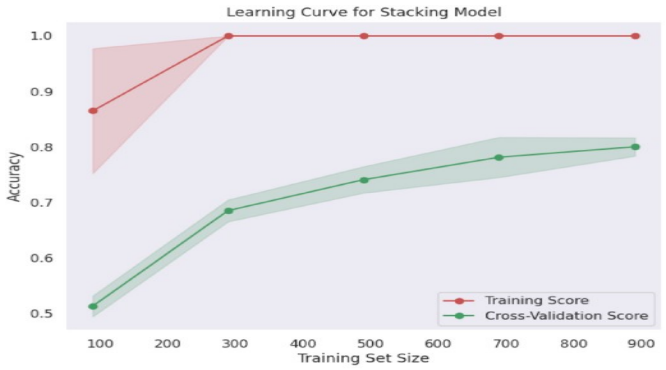
Fig. 7. 4x4 Confusion Matrix for ML classifier ((a) RF (b) KNN (c) GB (d) XGB (e) SVC (f) DT (g) LR) After Applying Scaling Techniques.

Table 7
Overall accuracy of each classifier before and after feature scaling.

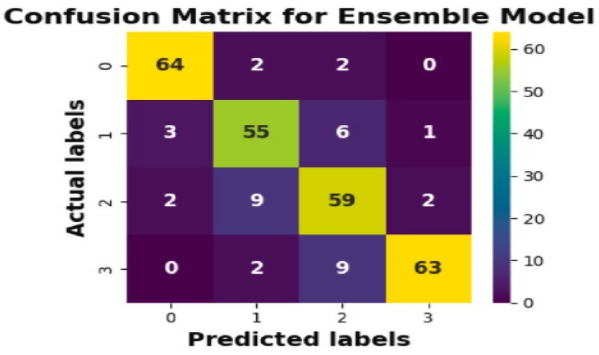
Classifiers	Before Scaling				After Scaling			
	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score	Accuracy
RF	**75.89 %**	**74.25 %**	**72.86 %**	**74.29 %**	**80.90 %**	**78.29 %**	**77.04 %**	**78.29 %**
KNN	73.84 %	73.14 %	71.59 %	73.14 %	72.90 %	72.57 %	72.17 %	72.57 %
GB	71.52 %	70.86 %	70.28 %	70.86 %	71.93 %	70.86 %	70.19 %	70.86 %
XGB	71.58 %	70.29 %	69.35 %	70.29 %	71.58 %	70.29 %	69.35 %	70.29 %
SVC	70.16 %	68.00 %	67.76 %	68.00 %	78.31 %	74.29 %	72.58 %	74.29 %
DT	58.38 %	57.71 %	57.46 %	57.71 %	60.33 %	60.57 %	60.35 %	60.57 %
LR	53.25 %	54.89 %	52.46 %	54.86 %	52.57 %	54.86 %	52.30 %	54.86 %

Table 8
Overall accuracy of each classifier after applying SMOTETomek and SMOTE-ENN.

Classifier	SMOTETomek				SMOTE-ENN			
	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score	Accuracy
RF	80.86 %	80.45 %	80.61 %	80.45 %	84.60 %	84.21 %	84.31 %	84.23 %
KNN	81.47 %	81.95 %	81.10 %	81.95 %	82.23 %	81.72 %	81.40 %	81.72 %
GB	84.91 %	84.59 %	84.66 %	**84.59 %**	85.51 %	85.30 %	85.31 %	**85.30 %**
XGB	81.25 %	80.83 %	80.93 %	80.83 %	83.17 %	83.15 %	83.11 %	83.15 %
SVC	85.66 %	84.59 %	84.94 %	84.59 %	85.23 %	84.59 %	84.77 %	84.59 %
DT	69.90 %	70.30 %	69.96 %	70.30 %	73.48 %	73.84 %	73.59 %	73.84 %
LR	56.10 %	56.39 %	56.20 %	56.39 %	59.06 %	58.78 %	58.91 %	58.78 %



(i): Learning Curve for Stacking Ensemble Model



(ii): 4x4 Confusion Matrix for Stacking Ensemble Model.

Fig. 8. Fig. 8 (i) Learning curve for stacking ensemble model (ii): 4x4 confusion matrix for stacking ensemble model.

Table 9
Class-wise estimated binary confusion matrix value for stacking ensemble model.

Classifiers	Class	TP	TN	FP	FN
Stacking Ensemble Model	Below Average	64	206	5	4
	Average	55	201	13	10
	Good	59	190	17	13
	Excellent	63	202	3	11

Table 10
Overall accuracy comparison of GB and stacking ensemble model.

Classifier	Precision	Recall	F1-score	Accuracy
GB	85.50 %	85.30 %	85.31 %	85.30 %
Stacking Ensemble Mode	86.80 %	86.37 %	86.49 %	**86.38 %**

$\tau(\zeta)]$ = mel's output without i^{th} feature

The output of the model can be written as a linear summation of SHAP base values and a constant base value as [44]:

$$e(z)' = k_0 + \sum_{i=1}^{n} k_i z' \tag{15}$$

Where, k_0 = base value, and $z' \epsilon \{0, 1\}$, $z' = 1$ when a feature is present or otherwise 0

3.5.10. *LIME (Local Interpretable Model-agnostic Explanations)*

LIME is another technique for interpreting ML models. This method generates localized interpretations of black-box models by training interpretable models on smaller samples of data. It explains the model's output for individual instances. LIME follows the following formula:

$$\gamma(x) = arg\ min\ L\ (f, g, \pi_x + \Omega(g), g \epsilon G \tag{16}$$

Where the loss function $\boldsymbol{L}$ is employed to measure the degree to which the understandable model $\boldsymbol{g}$ approximates the projection of the initial complex model $\boldsymbol{f}$.

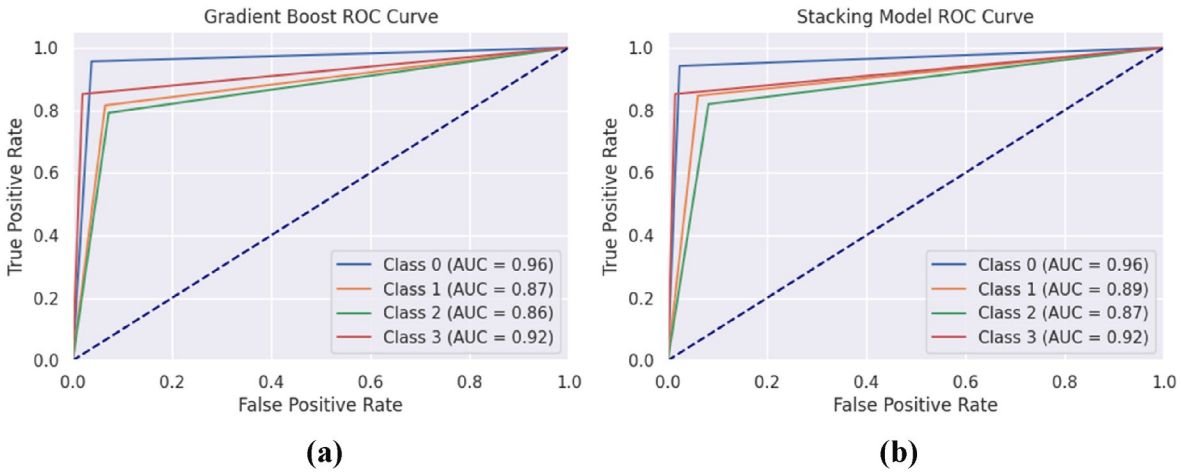


Fig. 9. Roc curve for (a) GB and (b) stacking ensemble model.

3.6. Performance evaluation metrics

For evaluating our models we have used four different types of performance evaluation metrics named Accuracy (ACC), Precision (PRE), Recall(REC), and F1-score(FS). To estimate the value of each evaluation metrics, the following equations are used.

$$\mathrm{ACC} = \frac{TP + TN}{TP + TN + FN + FP} \tag{17}$$

$$\mathrm{PRE} = \frac{TP}{TP + FP} \tag{18}$$

$$\mathrm{REC} = \frac{TP}{TP + FN} \tag{19}$$

$$\mathrm{FS} = 2 \times \frac{PRE \times REC}{PRE + REC} \tag{20}$$

Where,

TP = True Positive Rate.
TN = True Negative Rate.
FN= False Negative Rate.
FP= False Positive Rate.

4. Experimental result and discussion

The experimental setting that is used to predict student performance is described in this section. Based on various variables, ML classifiers may accurately estimate a student's performance. Google Collab was used to execute our proposed stacking ensemble model and the baseline classifiers. We programmed in Python and used several useful libraries, such as Scikit-learn, pandas, Seaborn, Matplotlib, and NumPy, for various tasks to accomplish this work. In addition, to train and test the models we relied on an "Intel® core (TM) i5″ CPU with 8 GB of RAM. Hyperparameter optimization techniques are used for both the baseline classifiers and the stacking ensemble model to achieve higher performance.

4.1. Impact of feature scaling on the performance of ML models

Firstly, we investigated the effect of feature scaling on the model's performance. After the completion of testing each classifier, a multiclass 4x4 performance matrix was generated for each which is shown in Figs. 6 and 7 and later converted into a 2x2 confusion matrix. equation (19) and (20) have been utilized to estimate the performance of each classifier.

Table 7 describes the overall performance of each classifier before and after applying feature scaling techniques. When the feature scaling technique is not applied, the performance evaluation metrics values of each classifier is very poor. The highest metrics values of Precision, Recall, F1-score, and Accuracy are 75.89 %, 74.25 %, 72.86 %, and 74.29 % respectively obtained by the Random Forest classifier. On the contrary, the lowest performance evaluation metrics values are found for the Logistic Regression classifier. After applying the feature scaling techniques, the performance of each classifier has been improved. The improved evaluation metrics values of Precision, Recall, F1-score, and Accuracy are 80.90 %, 78.29 %, 77.04 %, and 78.29 % respectively obtained by the Random Forest classifier.

4.2. Impact of SMOTE for data balancing

The SMOTE method was used to balance the data. A variety of SMOTE techniques exist and we have used SMOTETomek and SMOTE-ENN. Table 8 represents the overall performance for each classifier after applying SMOTETomek and SMOTE-ENN separately. The GB classifier achieved an accuracy for SMOTE-Tomek which is 84.59 % with a precision of 84.91 %, recall of 84.59 %, and F1-score of 84.66 %. While applying SMOTE-ENN the GB achieved an accuracy of 85.31 % with 85.51 %, 85.30 %, and 85.31 % of precision, recall, and F1-score respectively. In contrast, LR achieved the lowest accuracy of 56.39 % with a precision of 56.10 %, recall of 56.39 %, and F1-score of 56.20 %. But for SMOTE-ENN the accuracy is improved which is 58.78 %. Similarly, the other classifiers RF, KNN, XGB, and DT attained an accuracy of 80.45 %, 81.95 %, 80.83 %, 70.30 %, and 56.39 % respectively for SMOTETomek. However, after applying SMOTE-ENN it was increased to 84.23 %, 81.72 %, 83.15 %, 73.84 %, and 58.78 % respectively. Nonetheless, SVC has the save accuracy of 84.59 % for both SMOTE-ENN and SMOTETomek.

4.3. Ensemble classifier performance analysis

A custom stalking ensemble classifier has been utilized to train and test the balancing dataset. Fig. 8(i) shows the learning curve for the stalking ensemble model. It illustrates that there was an improvement in cross-validation accuracy as the training set size increased. The training accuracy remains consistently high but the cross-validation accuracy improves gradually which indicates enhanced generalization. Nevertheless, a performance disparity still exists which indicates certain overfitting tendencies may still exist even after SMOTE mitigates data imbalance. Therefore, to ensure the validity and robustness of our model multiple evaluation metrics were employed including Precision, Recall, and F1-Score which are crucial for imbalanced datasets. Additionally,

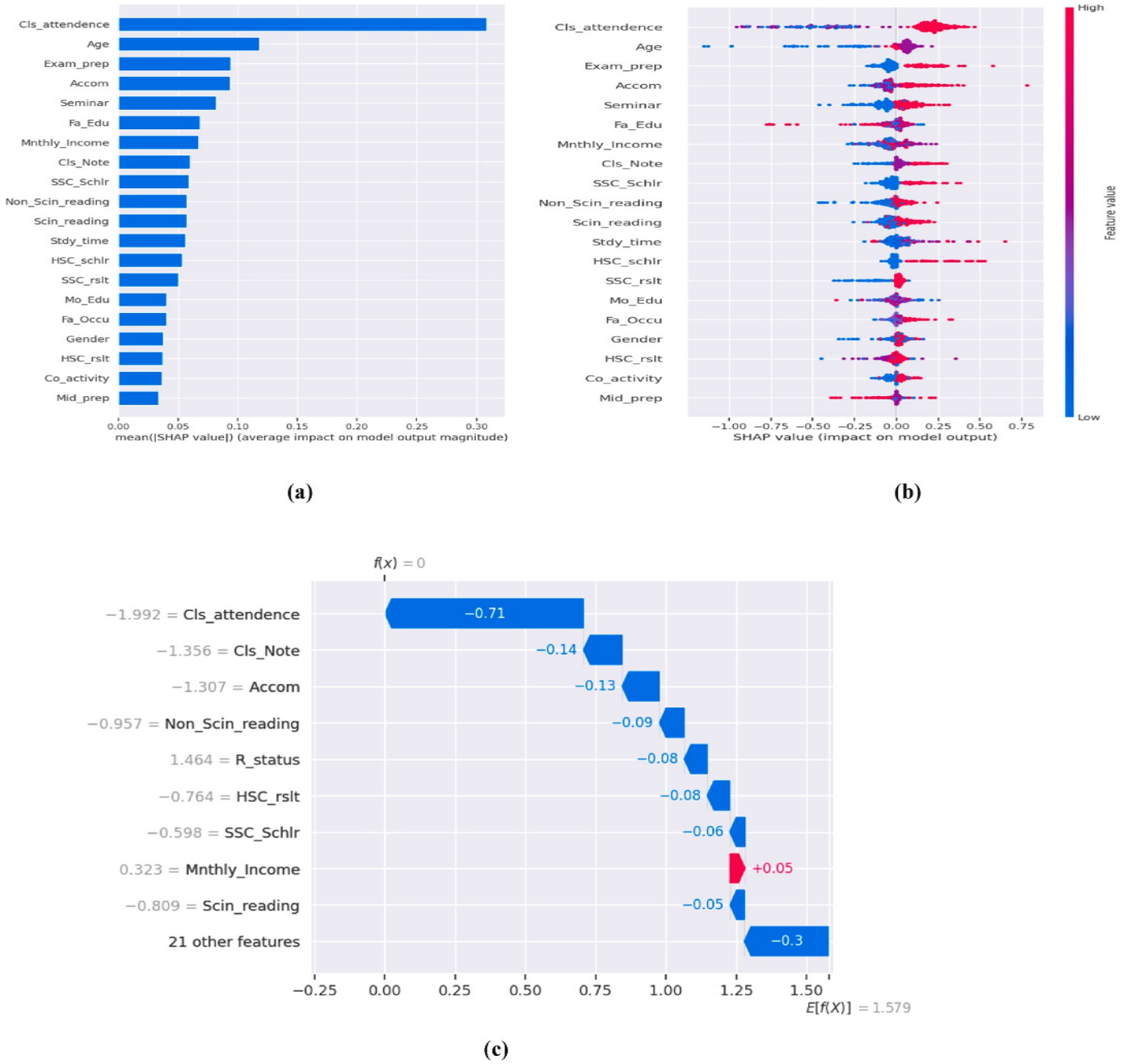


Fig. 10. (a) SHAP Value Counts for Each Feature; (b) Impact of SHAP Value on Model Output (Each class) (c) SHAP Waterfall Plot.

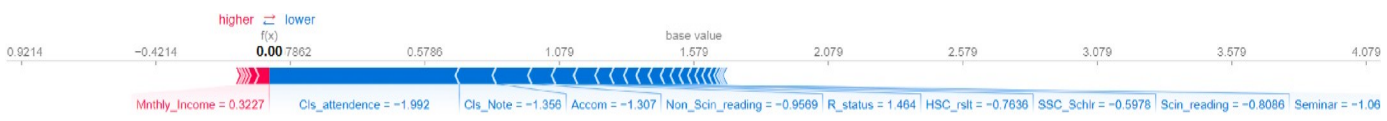


Fig. 11. Plotting of SHAP force plot for analyzing feature contribution.

we utilized ROC-AUC to measure the trade-off between sensitivity and specificity, providing a more comprehensive assessment of model performance. Furthermore, we integrated explainable AI techniques, namely SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations), to elucidate feature contributions and improve model transparency. The integration of these validation and interpretability techniques ensures that our model is both predictive and reliable, addressing key challenges associated with imbalanced datasets and overfitting. After the completion of the testing process, the obtained confusion matrix is presented in Fig. 8 (ii) and converted to the multiclass confusion matrix to binary format in Table 9.

Additionally, Table 10 illustrates the outperformed result of the stacking ensemble model with an accuracy of 86.38 %, and precision of 86.80 %, a recall of 86.37 %, F1-score of 86.49 %. In contrast, the GB classifier (best model after applying SMOTE) exhibits an accuracy of 85.30 % with 85.50 %, 85.30 %, and 85.31 % precision, recall, and F1-

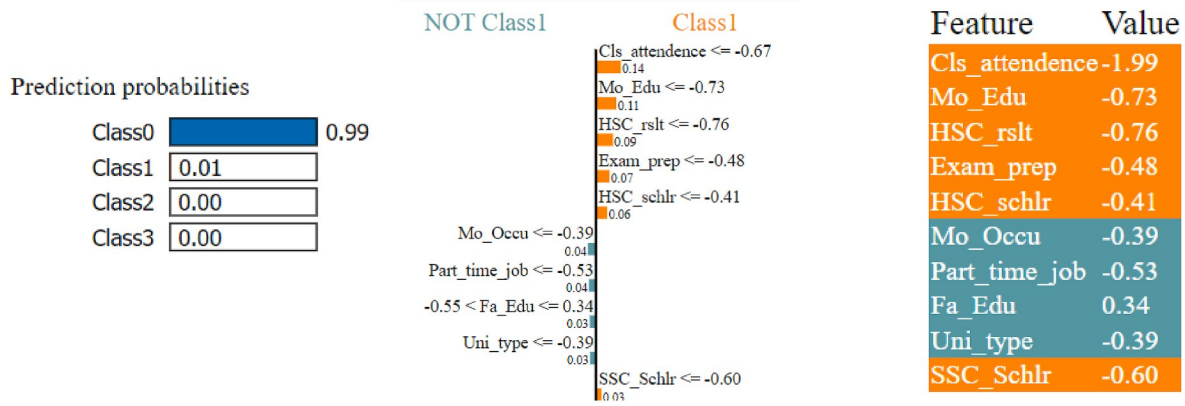


Fig. 12. Visualizing the stacking ensemble Model's output explanation (one instance) for LIME techniques.

Table 11
Comparative result analysis to predict student performance.

Reference	Sample Size	Number of feature	Data balancing	Highest Accuracy	Output Explanation
M. Zulfiker et al. [16]	400	8	x	Weighted Voting Classifier- 81.73 %	x
M. Anwar et al. [17]	518	17	x	ANN- 88 %	x
F. Janan et al. [18]	583	21	x	SVM- 81.25 %	x
M. Ashaduzzaman et al. [19]	687	07	x	SVM- 89.1262 %	x
K. Ahammad et al. [20]	400	31	x	MLP- 86.25 %	x
M. Hoque et al. [21]	850	14	x	J4-8- 85.6471	x
M. Mia et al. [22]	More than 1000	5	x	SVM- 85.76 %	x
This Study	872	**31**	✓	Stalking Ensemble classifier- 86.38 %	✓

score. It can be claimed that our custom model outperformed the state of art model.

The AUC values for each class are again highlighted in Fig. 9, which reveals that GB classifier received AUCs of 0.96, 0.87, 0.86, and 0.92 for the "Below Average," "Average," "Good," and "Excellent". In contrast, the suggested ensemble model fared better, with matching AUC values of 0.96, 0.87, 0.86, and 0.92.

4.4. Models interpretability analysis

As the highest accuracy has been achieved for our proposed stalking ensemble classifier, explanation of this classifier is analyzed applying SHAP and LIME techniques in order to find transparency. Both SHAP and LIME are crucial for enhancing the interpretability of our proposed Stacking Ensemble Model by providing both global and local explanations of its predictions. SHAP utilizes Shapley value aggregations and offers a fair and unbiased assessment of each feature's contribution by averaging over all possible orderings of feature inclusion. This allows us to identify the most influential features and understand their impact on model predictions through visualizations such as the summary plot, waterfall plot, and force plot. In contrast, LIME provides local interpretability by approximating the model's decision boundary for specific instances, highlighting the key features that influence individual predictions. Unlike SHAP, which provides a holistic view of feature importance across the dataset, LIME perturbs input data and trains an interpretable surrogate model to explain why a particular instance was classified into a specific category.

The SHAP summary plot combines the feature importance with feature effects on the model's prediction. Fig. 10(a) shows the bar plot for average SHAP values for each feature. This summary plot shows cls_attendence is the most crucial factor for model prediction, followed by age, Exam_prep and Accom. On the other hand, Fig. 10(b), the y-axis demonstrates the features such as cls_attendence, Age, Exam_prep, Accom, Seminar, Fa_Edu etc and the effect of these features are represented by the x-axis, which is the SHAP value. The color dots are used as the SHAP values where blue means negative influence and red means positive influence. According to the summary Plot High cls_attendence has a positive impact while low cls_attendence has a negative impact on the model output. Consequently, Fig. 10(c) shows how the features influence students' performance prediction from a base value of 1.579 to the final predicted value in the waterfall graph where the positive contribution is highlighted by red bars and the negative are represented by blue bars. The features Cls_attendence, cls_Note, Accom, Non_Scin_reading, R_status, HSC_rslt, SSC_Schlr, Scin_reading reduces the prediction by −0.71, −0.14, −0.13, −0.09, −0.08,-0.08, −0.06, and −0.809 respectively while only Mnthly_Income feature contributes positively.

Each feature's contribution to the model's prediction for a single instance is explained in depth via the force plot in SHAP. In Fig. 11 the base value is 1.579 whereas the arrows illustrate the contribution of each feature and red arrows mean higher contribution while blue means lower contribution. The red bar of Mnthly_Income increases the prediction by 0.3227 while cls_attendence, cls_Note, Accom decrease the prediction by 1.992, 1.356, 1.307 respectively.

We have used LIME to interpret the results of our ensemble model for a specific instance which is presented in Fig. 12. Unlike SHAP, which offers a global explanation for the whole dataset, LIME is used to explain the model's prediction for each particular occurrence. The leftmost section displays the prediction probability that exhibits the ensemble model predicted that the particular instance will be in the "Below Average" (Class0) category with 99 % confidence and with a very low probability for class1 ("Average") which is 1 %. For class2 ("Good") and class3 ("Excellent") the prediction probability is 0.00 indicating that they are almost impossible predictions for this data point. Furthermore, the Middle part yields the top ten features influencing the model's decision towards class0 and class 1. It demonstrates that the output class class 0 ("Below Average") is highly influenced by features like Mo_Occu, Part_time_job, Fa_Edu, and Uni_type. While cls_attendence, Mo_Edu, HSC_rslt, Exam_prep, HSC_schlr, and SSC_schlr characteristics assist the instance in avoiding falling within the class0 ("Below average") category. Furthermore, the actual values of these attributes are shown in the rightmost portions. These values represent the magnitude of the

feature's influence on the model's prediction for the corresponding class.

4.5. Comparison with existing studies

To prove the best effectiveness of our work, it is necessary to compare it with the relevant research. Table 11 presents a comprehensive comparison between the existing research and our findings. Most of the research was performed on a small dataset collected from a specific institution and there is no research on model explanations to predict student performance. Additionally, some research achieved the highest accuracy utilizing few features. Our research findings can be more effective in this context as we have utilized 872 data with 31 features from different institutions and obtained the highest accuracy (86.38 %) by applying a custom ensemble model incorporating XAI approaches such as LIME and SHAP to provide transparent and reliable predictions.

5. Conclusion and future work

This study investigated the factors that affect Bangladeshi undergraduate students' first semester academic performance, since it's a critical time in their educational journeys. The results of our study demonstrate that academic habits exert a substantial influence on educational success. Students who have a higher attendance rate demonstrate better academic performance compared to their peers with lower attendance. We have employed in-depth research to provide more accurate predictions with model explainability. We applied several data preprocessing mechanisms. In the first stage, we utilized various ML models to predict students' results. Then we compared the results of these models with our proposed stacking ensemble model. Our model performed better than these ML models and achieved an accuracy of 86.38 %. To make the prediction result more transparent and trustworthy, we have utilized SHAP and LIME which allows us to see the contribution of each variable to the prediction of students' results and help the practitioner, teachers, and policymakers in taking steps to improve students' academic performance.

In the future, we will expand our dataset by gathering data from a broader range of universities in Bangladesh as well as other countries to address generalizability issues. Additionally, we planned to customize the prediction process based on university types and departments. It will help us to understand the undergraduate educational landscape more comprehensively and suggest improved strategies for teachers, policymakers, and practitioners to help students.

CRediT authorship contribution statement

Fatema Tuz Johora: Writing – original draft, Resources, Methodology, Data curation, Conceptualization. **Md Nahid Hasan:** Writing – original draft, Resources, Methodology. **Aditya Rajbongshi:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **Md Ashrafuzzaman:** Writing – review & editing, Validation, Formal analysis. **Farzana Akter:** Writing – review & editing, Validation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] Chowdhury Mahfuzul Hoque, Absar Mir Mohammed Nurul, Quader Syed Manzur. Challenges and developments in the higher education system of Bangladesh: keys to way forward. In: Handbook of education systems in south asia; 2020. p. 1–32.

[2] Alamgir Zareen, et al. Enhancing student performance prediction via educational data mining on academic data. Inf Educ 2024;23.1:1–24.

[3] Zhou Qing, Quan Wenjun, Zhong Yu, Xiao Wei, Mou Chao, Wang Yong. Predicting high-risk students using Internet access logs. Knowl Inf Syst 2018;55:393–413.

[4] Aslam Nida, Khan Irfan, Alamri Leena, Almuslim Ranim. An improved early student's academic performance prediction using deep learning. International Journal of Emerging Technologies in Learning (iJET) 2021;16(12):108–22.

[5] Alsubhi Bashayer, Alharbi Basma, Aljojo Nahla, Banjar Ameen, Tashkandi Araek, Alghoson Abdullah, Al-Tirawi Anas. Effective feature prediction models for student performance. Eng Technol Appl Sci Res 2023;13(5):11937–44.

[6] Pallathadka Harikumar, Wenda Alex, Ramirez-Asís Edwin, Asís-López Maximiliano, Flores-Albornoz Judith, Phasinam Khongdet. Classification and prediction of student performance data using various machine learning algorithms. Materials today: proceedings 2023;80:3782–5.

[7] Ha Dinh Thi, Loan Pham Thi To, Nguyen Giap Cu, Huong Nguyen Thi Lien. An empirical study for student academic performance prediction using machine learning techniques. Int J Comput Sci Inf Secur 2020;18(3):75–82.

[8] Alyahyan Eyman, Düştegör Dilek. Predicting academic success in higher education: literature review and best practices. International Journal of Educational Technology in Higher Education 2020;17(1).

[9] Altaher A, BaRukab O. Prediction of student's academic performance based on adaptive neuro-fuzzy inference. Int. J. Comput. Sci. Netw. Secur. 2014;17(1): 165–9.

[10] Swamy Vinitra, et al. Evaluating the Explainers: black-box explainable machine learning for student success prediction in MOOCs. arXiv preprint arXiv:2207.00551 2022.

[11] Pereira Filipe Dwan, Fonseca Samuel C, Oliveira Elaine HT, Cristea Alexandra I, Bellhäuser Henrik, Rodrigues Luiz, Oliveira David BF, Isotani Seiji, Carvalho Leandro SG. Explaining individual and collective programming students' behavior by interpreting a black-box predictive model. IEEE Access 2021;9: 117097–119 [21].

[12] Chitti Manjari, Chitti Padmini, Jayabalan Manoj. Need for interpretable student performance prediction. In: 2020 13th international conference on developments in eSystems engineering (DeSE). IEEE; 2020. p. 269–72.

[13] Hoq Muntasir, Brusilovsky Peter, Akram Bita. Analysis of an explainable student performance prediction model in an introductory programming course. the 16th International Conference on Educational Data Mining (EDM 2023). 2023. International Educational Data Mining Society 2023.

[14] Pelegrina Guilherme Dean, Duarte Leonardo Tomazeli, Grabisch Michel. A k-additive Choquet integral-based approach to approximate the SHAP values for local interpretability in machine learning. Artif Intell 2023;325:104014.

[15] Raihan MJ, Khan MA, Kee SH, Nahid AA. Detection of the chronic kidney disease using XGBoost classifier and explaining the influence of the attributes on the model using SHAP. Sci Rep 2023;13:6263 [CrossRef].

[16] Zulfiker Md Sabab, Kabir Nasrin, Biswas Al Amin, Chakraborty Partha, Rahman Md Mahfujur. Predicting students' performance of the private universities of Bangladesh using machine learning approaches. Int J Adv Comput Sci Appl 2020; 11(3):672–9.

[17] Hossen Md Anwar, Bin Alamgir Rakib, Alam Arman Ul, Siddika Fatema, Hossain Shah Fahad, Arman Md Shohel. A web based four-tier architecture using reduced feature based neural network approach for prediction of student performance. In: 2021 2nd international conference on robotics, electrical and signal processing techniques (ICREST). IEEE; 2021. p. 269–73.

[18] Janan Farhatul, Kumar Ghosh Sourav. Prediction of student's performance using support vector machine classifier. In: Proc. Int. Conf. Ind. Eng. Oper. Manag; 2021. p. 7078–88.

[19] Pritom, Ahmed Iqbal "An analysis of student's academic record using data mining techniques and identification of key factors to aid student's performance.".

[20] Ahammad Khalil, Chakraborty Partha, Akter Evana, Honey Fomey Umme, Rahman Saifur. A comparative study of different machine learning techniques to predict the result of an individual student using previous performances. Int J Comput Sci Inf Secur 2021;19(1).

[21] Hoque Md Imdadul, Kalam Azad Abul, Tuhin Mohammad Abu Hurayra, Us Salehin Zayed. University students result analysis and prediction system by decision tree algorithm. Adv Sci Technol Eng Syst J 2020;5(3):115–22.

[22] Mia Md Jueal, Biswas AA, Sattar A, Habib Md T. Registration status prediction of students using machine learning in the context of Private University of Bangladesh. Int J Innovative Technol Explor Eng 2019;9(1):2594–600.

[23] Livieris Ioannis E, Karacapilidis Nikos, Domalis Georgios, Tsakalidis Dimitris. An advanced explainable and interpretable ML-based framework for educational data mining. In: International conference in methodologies and intelligent systems for techhnology enhanced learning. Cham: Springer Nature Switzerland; 2023. p. 87–96.

[24] Alwarthan Sarah, Aslam Nida, Ullah Khan Irfan. An explainable model for identifying at-risk student at higher education. IEEE Access 2022;10:107649–68.

[25] Afzaal Muhammad, Zia Aayesha, Nouri Jalal, Fors Uno. Informative feedback and explainable AI-based recommendations to support students' self-regulation. Technol Knowl Learn 2024;29(1):331–54.

[26] Nagy Marcell, Molontay Roland. Interpretable dropout prediction: towards XAI-based personalized intervention. Int J Artif Intell Educ 2023:1–27.

[27] Chen Hsing-Chung, Prasetyo Eko, Tseng Shian-Shyong, Putra Karisma Trinanda, Kusumawardani Sri Suning, Weng Chien-Erh. Week-wise student performance early prediction in virtual learning environment using a deep explainable artificial intelligence. Appl Sci 2022;12(4):1885.
[28] Adnan Muhammad, Irfan Uddin M, Khan Emel, Alharithi Fahd S, Amin Samina, Alzahrani Ahmad A. Earliest possible global and local interpretation of students' performance in virtual learning environment by leveraging explainable AI. IEEE Access 2022;10:129843–64.
[29] Khanna VV, Chadaga K, Sampathila N, Prabhu S, Bhandage V, Hegde GK. A distinctive explainable machine learning framework for detection of polycystic ovary syndrome. Appl. Syst. Innov. 2023;6:32 [CrossRef].
[30] Han Hui, Wang Wen-Yuan, Mao Bing-Huan. Borderline-SMOTE: a new over-sampling method in imbalanced data sets learning. In: International conference on intelligent computing. Berlin, Heidelberg: Springer Berlin Heidelberg; 2005. p. 878–87.
[31] scikit-learn-contrib, Imbalance learn. "SMOTEN." July 19, 2024. https://imbalanced-learn.org/stable/references/generated/imblearn.over_sampling.SMOTEN.
[32] scikit-learn-contrib, Imbalanced learn. "SMOTETomek." July 20, 2024. https://imbalanced-learn.org/stable/references/generated/imblearn.combine.SMOTETomek.
[33] Ozsahin Dilber Uzun, Taiwo Mustapha Mubarak, Saleh Mubarak Auwalu, Ameen Zubaida Said, Uzun Berna. Impact of feature scaling on machine learning models for the diagnosis of diabetes. In: 2022 international conference on artificial intelligence in everything (AIE). IEEE; 2022. p. 87–94.
[34] Ha Dinh Thi, Loan Pham Thi To, Nguyen Giap Cu, Huong Nguyen Thi Lien. An empirical study for student academic performance prediction using machine learning techniques. Int J Comput Sci Inf Secur 2020;18(3):75–82.
[35] Genuer Robin, Poggi Jean-Michel, Genuer Robin, Poggi Jean-Michel. Random forests. Springer International Publishing; 2020.
[36] Ali Zeravan Arif, Abduljabbar Ziyad H, Taher Hanan A, Sallow Amira Bibo, Almufti Saman M. Exploring the power of eXtreme gradient boosting algorithm in machine learning: a review. Academic Journal of Nawroz University 2023;12(2):320–34.
[37] Chen Tianqi, Tong He, Benesty Michael, Khotilovich Vadim, Yuan Tang, Cho Hyunsu, Chen Kailong, Mitchell Rory, Cano Ignacio, Zhou Tianyi. Xgboost: extreme gradient boosting. R package version 0 2015;1(4):1–4. 4-2.
[38] Christopher. Antony. "K-Nearest Neighbor." Medium 2024;21. https://medium.com/swlh/k-nearest-neighbor-ca2593d7a3c4. July.
[39] Alzubi Jafar, Nayyar Anand, Kumar Akshi. Machine learning from theory to algorithms: an overview. In: Journal of physics: conference series. vol. 2. IOP Publishing; 2018, 012012. vol. 114.
[40] Abdullayeva Fargana J. Advanced persistent threat attack detection method in cloud computing based on autoencoder and softmax regression algorithm. Array 2021;10:100067.
[41] Pes Barbara. Learning from high-dimensional and class-imbalanced datasets using random forests. Information 2021;12(8):286.
[42] Lyashevska Olga, Malone Fiona, MacCarthy Eugene, Fiehler Jens, Buhk Jan-Hendrik, Morris Liam. Class imbalance in gradient boosting classification algorithms: application to experimental stroke data. Stat Methods Med Res 2021;30(3):916–25.
[43] Lundberg SM, Lee S-I. A unified approach to interpreting model predictions. Adv Neural Inf Process Syst 2017;30:4765–74.
[44] Khattak Afaq, Chan Pak-Wai, Chen Feng, Peng Haorong. Prediction and interpretation of low-level wind shear criticality based on its altitude above runway level: application of Bayesian optimization–ensemble learning classifiers and SHapley additive exPlanations. Atmosphere 2022;13(12):2102.